MILITARY SPECIFICATION

CARTRIDGE, 40MM, WHITE SMOKE, CANOPY, M680

METAL PARTS FOR AND LOADING, ASSEMBLING AND PACKING

This specification is approved for use by the U.S. Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification contains requirements not normally covered by the drawings and provides quality assurance provisions for the fabrication of parts and loading, assembling and packing for Cartridge, 40MM, White Smoke, Canopy, M680.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. - The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-223	- Black Powder
MIL-C-5541	- Chemical Films and Chemical Film Materials for Aluminum and Aluminum Alloys
MIL-A-48078	- Ammunition, Standard Quality Assurance Provisions, General Specification For

FSC: 1310

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-331 - Fuze and Fuze Components, Environmental and Performance Tests for
MIL-STD-1169- Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1234- Pyrotechnics Sampling, Inspection and Testing

DRAWINGS (See 6.8)

U.S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

9317512 - Cartridge Assembly, 40MM, White Smoke, Canopy, M680
9209204 - Box, Ammo, Metal for Cartridge, 40MM, White Star, Cluster or Smoke Canopy or Parachute
9209205 - Box, Wirebound, Packing, Ammunition for Cartridge, 40MM, White Star, Cluster or Smoke Canopy or Parachute

INSPECTION EQUIPMENT DRAWINGS

9202770 - Chamber Gage
9202783 - Alignment Check
8827895 - Limit Check

2.2 Other Publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

TITLE 49 - Transportation, Parts 100-199

(The Code of Federal Regulations is available from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402. Orders should specify "49 CFR 100-199 (latest revision)").

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AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM B 117-73 - Standard Method of Salt Spray (Fog) Testing

(Application for copies should be addressed to: American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103)

3. REQUIREMENTS

3.1 Material. - Materials shall be in accordance with the applicable drawings and specifications.

3.2 Cartridge. - The cartridge shall comply with all requirements specified on Drawing (Dwg.) 9317512, all associated drawings, and with all requirements specified in applicable specificaitons.

3.3 Transportation vibration. - The cartridge shall comply with the following requirements:

3.3.1 The cartridge shall be safe to transport following this test (See 6.5).

3.3.2 There shall be no evidence of external damage to the cartridge that will affect the intended function (See 6.6).

3.4 Functioning. - The cartridge shall comply withthe following requirements;

3.4.1 Altitude. - The projectile shall function at an average altitude of 300 feet plus (+) 50 feet or minus (-) 25 feet with no item below 200 feet.

3.4.2 Burning of smoke assembly. - The smoke assembly shall ignite and emit smoke for 100 ± 20 seconds while suspended fromthe parachute at ambient temperature.

3.4.3 Parachute. - The parachute shall not separate from the smoke assembly at ejection or delay opening from time of ejection for more than 5 seconds.

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3.4.4 Firing. - No projectile shall stick in the gun bore and the projectile shall not burst or eject the smoke assembly 50 feet from the launcher.

3.4.5 Hot Temperature (160 defrees F. plus or minus 5 degrees F). - The cartridge shall comply with the requirements of 3.4.1, 3.4.2, 3.4.3 and 3.4.4.

3.4.6 Ambient temperature (70 degrees F. plus or minus 5 degrees F). - The cartridge shall comply with the requirements of 3.4.1, 3.4.2, 3.4.3 and 3.4.4.

3.4.7 Cold temperature (-65 degrees F. plus or minus 5 degrees F). - The cartridge shall comply withthe requirements of 3.4.1, 3.4.2, 3.4.3 and 3.4.4.

3.5 First article inspection. - This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.6 Workmanship. - All parts and assemblies shall be fabricated, loaded and assembled in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, grease, rust and other foreign matter. The cleaning method used shall not be injurious to any parts, nor shall the parts be contaminated by the cleaning agents. Exterior surface coatings shall be continuous; however, a few light scratches not exposing base material may be permitted. All required marking and stampling shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. - Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specificaiton.

4.2 Classification of inspections. - The following types of inspection shall be conducted on this item:

a. First Article Inspection
b. Quality Conformance Inspection

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4.3 First article inspection. -

4.3.1 Submission. - The contractor shall submit a first article as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Description	Drawing	Quantity
Body	9243900	25 (5 painted)
Disc anchor	9252374	25
Ogive	9317516	25
Smoke canisters	9252377	25
Disc, ejection	9252397	25
Case, Smoke	9255798	25
Housing delay	9252378	25 (5 for salt spray)
Parachute assembly	9252371	25
Smoke canister loading assembly	9252368	25
Smoke assembly	9252368	25
Delay loading assembly	9252369	25
Cartridges	9317512	465(4440 to a Government Proving Ground for functioning tests)

4.3.2 Inspections to be performed. - See MIL-A-48078 and Table I herein.

4.3.3 Rejection. - See MIL-A-48078 and the following;

4.3.3.1 Proving Ground Tests

a. Any critical defect occurs in any phase (Table III).

b. Average altitude of the total sample fails the applicable requirement (Table III and 3.4.1).

c. A total of eight (8) or more in the hot phase fail any of the following:

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(1) Parachute failure (3.4.3).

(2) Item fails to function (Table III).

(3) Altitude less than 200 ft (Table III). Burning time shall be recorded for informational purposes (average) (3.4.2).

d. A total of twenty-two (22) or more in the ambient phase fail any of the following;

(1) Parachute failure (3.4.3).

(2) Item fails to function (Table (III).

(3) Altitude less than 200 ft (Table III).

(4) Burning time incorrect (average) (3.4.2).

e. A total of eight (8) or more in the cold phase fail any of the following:

(1) Parachute failure (3.4.3).

(2) Item fails to function (Table III).

(3) Altitude less than 200 ft (Table III). Burning time shall be recorded for informational purposes (average) (3.4.2).

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63114(AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
	Cartridge and components				See below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Body (Dwg. 9243900)				
	Prior to painting				
	Examination for defects	20		3.2	4.4.2.1
	Salt spray	5		3.2	4.5.4
	After painting				
	Examination for defects	5		3.2	4.4.2.2
	Ogive (Dwg. 9317510)				
	Examination for defects	25		3.2	4.4.2.3
	Delay carrier (Dwg. 9252378)				
	Examination for defects	20		3.2	4.4.2.4
	Salt spray	5		3.2	4.5.4
	Disc Anchor (Dwg. 9252374)				
	Examination for defects	25		3.2	4.4.2.5
	Canister Smoke				
	(Dwg. 9252377)				
	Examination for defects	25		3.2	4.4.2.6
	Disc, Ejection				
	(Dwg. 9252367)				
	Examination for defects	25		3.2	4.4.2.7

NOTES:

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63114(AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
	Cartridge and components				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Smoke Canister Loading Assembly (Dwg.9255707)				
	Examination for defects	25		3.2	4.4.2.9
	Burn time	25		3.2	4.5.5
	Parachute Assembly (Dwg. 9252371)				
	Examination for defects	25		3.2	4.4.2.10
	Delay Loading Assembly (Dwg. 9252369)				
	Burn Time	25		3.2	4.5.4
	Smoke Assembly (Dwg. (252368)				
	Examination for defects	25		3.2	4.4.2.11
	Cartridge (Dwg. 9317512)				
	Examination for defects	25		3.2	4.4.2.15
	Pull test of projectile	25		3.2	4.5.7
	*Transportation-vibration	440		3.2	4.5.8
	*Air pressure	440		3.2	4.5.9
	Functioning				
	Hot	80		3.2	4.5.10.1
	Ambient	280		3.2	4.5.10.2
	Cold	80		3.2	4.5.10.3

NOTES: *Prior to submission of the 440 samples to a Gov't proving ground for functioning tests, the samples shall be inspected at the contractors plant for these tests.

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4.4 Quality conformance inspection. -

4.4.1 Inspection lot formation. - Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, each inspection lot shall contain:

a. Projectile metal parts from one interfix lot number from one manufacturer.
b. Delay composition from not more than one lot number.
c. Smoke composition produced by one manufacturer under one continuous set of operating conditions and which consist of one or more batches that have been subjected to the same unit chemical or physical mixing process intended to make the final product homogeneous.
d. Cartridge case loaded assemblies from one interfix lot number from one manufacturer.
e. Smoke assemblies from one interfix lot number from one manufacturer.
f. Ignition composition from not more than one lot number.
g. Delay Assemblies from one interfix lot number from one manufacturer.
g. Disc ejection composition from not more than one lot number

4.4.2 Examination. - See MIL-A-48078.

a. Sampling Plans. - Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63114(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.1	Body, prior to Painting		9243909
			NEXT HIGHER ASSEMBLY
			9317514

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Largest outside diameter		0.40%	3.2	Gage
102	Concentricity of largest outside diameter with largest outside diameter		0.40%	3.2	Gage
103	Second largest outside diameter		0.40%	3.2	Gage
104	Concentricity of second largest outside diameter with smallest inside diameter		0.40%	3.2	Gage
105	Concentricity of largest inside diameter with smallest inside diameter		0.40%	3.2	Gage
106	Salt Spray	*	*	3.2	4.5.3
107	Pitch diameter of thread		0.40%	3.2	Gage
108	Minor diameter of thread, max.		0.40%	3.2	Gage
109	Perpendicularity of open end with smallest inside diameter		0.40%	3.2	Gage
110	Metal defective		0.40%	3.5 & 6.2	Visual
Minor					
201	Smallest outside diameter		0.65%	3.2	Gage
202	Groove diameter		0.65%	3.2	Gage
203	Width of rotating bands		0.65%	3.2	Gage
204	Total length		0.65%	3.2	Gage
205	Width of flange		0.65%	3.2	Gage
206	Distance to rotating band		0.65%	3.2	Gage
207	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES: *Sampling and rejection shall be in accordance with MIL-C-5541.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63114(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9243000
4.4.2.2	Body after Painting				NEXT HIGHER ASSEMBLY 9317514
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Second largest outside diameter		0.40%	3.2	Gage
Minor 201	Color incorrect		0.65%	3.2	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63114(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9317516
4.4.2.3	Ogive				NEXT HIGHER ASSEMBLY 9317514
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Incorrect color		100% (See 4.5.1)	3.2	Visual
Major					
101	Large outside diameter		0.40%	3.2	Gage
102	Concentricity of large outside diameter with third largest outside diameter		0.40%	3.2	Gage
103	Second largest outside diameter		0.40%	3.2	Gage
104	Inside diameter		0.40%	3.2	Gage
Minor					
201	Total length		0.65%	3.2	Gage
202	Width of slots		0.65%	3.2	Gage
203	Diameter of groove		0.65%	3.2	Gage
204	Width of groove		0.65%	3.2	Gage
205	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-83114(AP)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9252378
4.4.2.4	Relay Carrier				NEXT HIGHER ASSEMBLY 9252369
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Salt Spray	*	*	3.2	4.5.4
102	Pitch diameter of thread		0.40%	3.2	Gage
103	Major diameter of thread		0.40%	3.2	Gage
104	Small inside diameter		0.40%	3.2	Gage
105	Concentricity of small inside diameter with large inside diameter		0.40%	3.2	Gage
106	Large inside diameter		0.40%	3.2	Gage
Minor					
201	Length to flange		0.65%	3.2	Gage
202	Concentricity of body diameter with large inside diameter		0.65%	3.2	Gage
203	Diameter of thread undercut		0.65%	3.2	Gage
204	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES: *Sampling and rejection shall be in accordance with MIL-C-5541.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63114(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9252374
4.4.2.5	Disc Anchor				NEXT HIGHER ASSEMBLY 9252376
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
Critical	None defined		(See 4.5.2)		
Major					
101	Outside diameter		0.40%	3.2	Gage
102	Thickness		0.40%	3.2	Gage
Minor					
201	Hole diameter		0.65%	3.2	Gage
202	Evidence of poor workmanship		0.65%	3.6	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63114(AD)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.6	Canister Smoke				9252377
					NEXT HIGHER ASSEMBLY
					9255797
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined		(See 4.5.2)		
Major					
101	Inside diameter		0.4[illegible]	3.2	Gage
102	Hole diameter		0.40%	3.2	Gage
Minor					
201	Body wall thickness		0.65%	3.2	Gage
[illegible]	Thickness through bottom		0.65%	3.2	Gage
[illegible]	Total length		0.65%	3.2	Gage
[illegible]	Outside radius, hole end		0.65%	3.2	Gage
205	Depth of hole (alternate design)		0.65%	3.2	Gage
206	Location of holes		0.65%	3.2	Gage
207	Incorrect number of holes		0.65%	3.2	Visual
208	Evidence of poor workmanship		0.65%	3.6	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63114(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.7	Disc, Ejection				9252397
					NEXT HIGHER ASSEMBLY
					9252390
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined		(See 4.5.1)		
Major					
101	Outside diameter		0.40%	3.2	Gage
102	Diameter of hole		0.40%	3.2	Gage
103	Depth of cavity		0.40%	3.2	Gage
Minor					
201	Total length		0.65%	3.2	Gage
202	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-83174(AS)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.2	Grate canister loading assembly, prior to loading				8888707
					NEXT HIGHER ASSEMBLY
					8888888
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Wire mesh disc missing		0.40%	3.2	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63114(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.9	Smoke canister loading assembly		9255797
			NEXT HIGHER ASSEMBLY 9252368

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
~~101~~	Depth to composition		0.40%	3.2	Case
102	Static burn time	32	1.5%	3.2	4.5.5
Minor					
~~201~~	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63114 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.10	Parachute Assembly	1 OF 1	9252371
			NEXT HIGHER ASSEMBLY 9317514

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Tears or holes in parachute		0.40%	3.2	Visual
102	Shroud lines not attached to parachute		0.40%	3.2	Visual
103	Shroud lines tangled		0.40%	3.2	Visual
104	Coupling not secure to shroud lines		0.40%	3.2	Visual

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CLASSIFICATION OF DEFECTS & TESTS MIL-C-63114 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.11	Smoke Assembly				9252368
					NEXT HIGHER ASSEMBLY
					9317514
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter		0.40%	3.2	Gage
102	Roll over crimp not full 360°		0.40%	3.2	Visual
Minor					
201	Total length		0.65%	3.2	Gage
202	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63114 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.12	Delay Assembly				9252369
					NEXT HIGHER ASSEMBLY 9317514
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Static burn time	32	1.5%	3.2	4.5.4

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63114 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.13	Projectile assembly, prior to assembling smoke assembly				9317514
					NEXT HIGHER ASSEMBLY
					9317512
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Coupling of smoke assembly not secure to chain		0.40%	3.2	Visual
102	Spacer missing		0.40%	3.2	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63114 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9317514
4.4.2.14	Projectile assembly, prior to assembling ogive				NEXT HIGHER ASSEMBLY 9317512
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Parachute missing		0.40%	3.2	Visual
102	O-ring missing		0.40%	3.2	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63114 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9317514
4.4.2.15	Projectile Assembly				NEXT HIGHER ASSEMBLY 9317512
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Gap present between delay assembly and body		0.40%	3.2	Visual
102	Color incorrect		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63114 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9317512
4.4.2.16	Cartridge				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Chamber gage failure		100%	3.2	9202770 9202783 8827895
2	Incorrect ogive		100%	3.2	Visual
3	Marking incorrect		100%	3.2	Visual
Major					
101	Marking unidentifiable		0.40%	3.2	Visual
102	Pull test	50	3.50%	3.2	4.5.7
103	Transportation-vibration			3.3	4.4.3.3, 4.4.3.6 and 4.5.8
104	Air pressure			3.2	4.4.3.2, 4.4.3.4 4.4.3.7 and 4.5.9
105	Functioning			3.4	4.4.3.5, 4.4.3.8 and 4.5.10
Minor					
201	Total length		0.65%	3.2	Gage
202	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63114 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9209204
4.4.2.17	Unsealed Metal Box				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Incorrect number of cartridges		0.40%	3.2	Visual
102	Desiccant bags missing		0.40%	3.2	Visual
103	Rubber gasket damaged or improperly assembled		0.40%	3.2	Visual
Minor					
201	Fillers missing		0.65%	3.2	Visual
202	Tube missing		0.65%	3.2	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63114 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.18	Sealed Metal Box				9209204
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Air pressure	*	*	32.	4.5.11
Minor					
201	Marking misleading or unidentifiable		0.65%	3.2	Visual
202	Contents loose		0.65%	3.2	Manual

NOTES: *One (1) packed, sealed box shall be selected from each twenty (20) boxes packed. If one (1) box fails, the quantity represented by the sample shall be rejected.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-C-63114 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.19	Sealed Wooden Packing Box		9209205
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Minor					
201	Marking misleading or unidentifiable		0.65%	3.2	Visual
202	Contents loose		0.65%	3.2	Manual
203	ICC nomenclature missing		0.65%	3.2	Visual

NOTES:

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4.4.3 Testing. -

4.4.3.1 Moisture Content (See Table 1), Major -

TABLE I

Material	Drawing
Delay Composition	9252369
Ignition Composition	9255797
Composition Ejection Disc	9252399
Smoke Composition	9255797

The contractor shall provide adequate controls to insure compliance with the requirements and shall test for verification at least one (1) sample of each material from each eight hour's production of cartridges. Composite samples shall not be used. If the moisture content of a sample exceeds the requirement and loading has not begun, that quantity of material represented by the sample shall be rejected. If assemblies have been loaded with material containing excessive moisture, the remaining material that is represented by the sample together with all ccartridges loaded with the non-forming material shall be rejected.

4.4.3.2 Air pressure. - This test shall be conducted 100 percent. Any cartridge which fails to comply with the requirement specified on the applicable drawing shall be classified defective and removed from the lot.

4.4.3.3 Transportation-vibration. - After completion of the 100 percent air pressure test and beginning with the first lot produced and continuing until three (3) acceptable lots (lots not accepted by disposition) have complied with the acceptance criteria specified in Table II, two-hundred and sixty-four (264) cartridges shall be selected from each lot, packed in their regular shipping containers and subjected to this test. The cartridges shall be observed and examined visually without disassembly for any evidence of failure to comply with the requirements of Table II.

TABLE II

Defect	Drawing
Cartridge not safe to transport following test	Critical
Cartridge damaged after test	Major

The lot shall be rejected if any defect as classified in Table II occurs.

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4.4.3.4 Air pressure. - After completion of the transportation-vibration test, the two-hundred and sixty-four (264) samples from each lot shall be subjected to this test. If nine (9) or more cartridges fail to comply with the requirement specified on the applicable drawing, the lot shall be rejected. Cartridges which fail this test shall be discarded and replaced with units which have been successfully subjected to transportation-vibration test prior to subsequent testing.

4.4.3.5 Functioning. - After completion of the air pressure test and beginning with the first lot produced and continuing until three (3) acceptable lots (lots not accepted by disposition) have complied with the criteria of Table III, the two-hundred and sixty-four (264) samples from each lot shall be packed and shipped to a government proving gorund and tested as follows:

4.4.3.5.1 Hot - Forty-four (44) samples shall be subjected to this test.

4.4.3.5.2 Ambient - One-hundred and seventy-six (176) samples shall be subjected to this test.

4.4.3.5.3 Cold - Forty-four (44) samples shall be subjected to this test.

4.4.3.5.4 Rejection. - The lot shall be rejected if;

a. Any critical defects occur in any phase (See Table III).

b. Average altitude of the total sample fails the applicable requirement (See 3.4.1).

c. A total of six (6) or more in the hot phase fail any of the following:

(1) Parachute failure (See 3.4.3).

(2) Item fails to function (See Table III).

(3) Altitude less than 200 ft (Table III). Burning time shall be recorded for informational purposes (average).

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d. A total of fifteen (15) or more in the ambient phase fail any of the following:

(1) Parachute failure (3.4.3).

(2) Item fails to function (Table III).

(3) Altitude less than 200 ft (Table III).

(4) Burning time incorrect (average) (3.4.2).

e. A total of six (6) or more in the cold phase fail any of the following;

(1) Parachute failure (3.4.3).

(2) Item fails to functiong (Table III).

(3) Altitude less than 200 ft (Table III). Burning time shall be recorded for informational purposes (average) (3.4.2).

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TABLE III

Defect	Classification
Any projectile sticks in the gun bore	Critical
Incorrect smoke color	Critical
Burst or ejects 50 ft from the launcher	Critical
Burning time incorrect (average) ambient only	Major
Average altitude	Major
Parachute failure	Major
Altitude less than 200 feet	Major
Item fails to function	Major

4.4.3.6 Transportation-vibration, regular production - After three (3) acceptable lots have complied with 4.4.3.3 and 4.4.3.5, one-hundred and thirty-two (132) of the samples subjected to the 100% air pressure test shall be selected from each lot, packed in their regular shipping containers and subjected to this test. The lot shall be rejected if any defect as classified in Table II occurs.

4.4.3.7 Air pressure, regular production. - After completion of the transportation-vibration test the one-hundred and thirty two (132) samples shall be removed from their shipping containers and subjected to this test. If eight (8) or more samples fail to comply with the requirement specified on the applicable drawing, the lot shall be rejected. Cartridges which fail this test shall be discarded and replaced with units which have been successfully subjected to transportation-vibration test prior to subsequent testing.

4.4.3.8 Functioning, regular production - After three (3) acceptable lots have complied with 4.4.3.5, the one-hundred and thirty-two (132) samples subjected to the air pressure test shall be packed and shipped to a Government proving ground and tested as follows:

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4.4.3.8.1 Hot - Twenty-six (26) samples shall be subjected to this test.

4.4.3.8.2 Ambient - Eighty (80) samples shall be subjected to this test.

4.4.3.8.3 Cold - Twenty-six (26) samples shall be subjected to this test.

4.4.3.8.4 Rejection - The lot shall be rejected if:

a. Any critical defect occurs in any phase (See Table III).

b. Average altitude if the total sample fails the applicable requirement (See 3.4.1).

c. A total of four (4) or more in the hot phase fail any of the following:

(1) Parachute failure (See 3.4.3).

(2) Item fails to function(See Table III).

(3) Altitude less than 200 ft (Table III). Burning time shall be recorded for informational purposes (average) (3.4.2).

d. A total of eight (8) or more in the ambient phase fail any of the following;

(1) Parachute failure (3.4.3).

(2) Item fails to functiong (Table III).

(3) Altitude less than 200 ft (Table III).

(4) Burning time incorrect (average) (3.4.2).

e. A total of four (4) or more in the cold phase fail any of the following;

(1) Parachute failure (3.4.3).

(2) Item fails to function (Table III).

(3) Altitude less than 200 ft (Table III). Burning time shall be recorded for informational purposes (average) (3.4.2).

If any lot fails, the consecutive lots thereafter shall revert to the criteria of 4.4.3.3.

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4.4.3.9 Suitability test plan (information test only) - Beginning with the first lot produced by each producer and continuing until the producer has submitted three (3) acceptable lots (lots not accepted by disposition) that are accepted in accordance with 4.4.3.5.1, 4.4.3.5.2 and 4.4.3.5.3, an extra sample of eighty-eight (88) rounds shall be selected from each lot and preconditioned and tested as follows:

a. The eighty-eight (88) rounds shall be packaged in their regular shipping containers and subjected to transportation vibration environmental in accordance with MIL-STD-331, Test NO. 104, Procedure No. II.

b. After transportation vibration, the rounds shall be unpacked and subjected to the JAN temperature and humidity cycle (14 days) specified in MIL-STD-331, Test No. 105.

c. After temperature and humidity cycle, the rounds shall be temperature conditioned and fired as follows:

30 rounds + 165 degrees F. ± 5 degrees F.
28 rounds ambient (between ∓ 40 and + 110 degrees F.)
30 rounds - 65 degrees F. ± 5 degrees F.

d. Fire the rounds in accordance with paragraph 4.5.10 and with TECOM Material Test Procedure, TECP 700-700 Volume 4. Test results, failure analysis reports and other information as requested shall be forwarded to ARRADCOM, ATTN: DRDAR-QA-A-Q.

4.4.4 Inspection equipment. - The inspection equipment required to perform the examinations and test prescribed herein is described in the Paragraph Reference/Inspection Method column in the tables starting with Paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of mIL-A-48078 and 6.3 herein.

4.5 Test methods and procedures

4.5.1 Dimensional control of molded or plastic parts - In place of the normal sampling and inspections associated with the Classification of Defects, and after a curing time

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(3) parts (as molded) from each cavity shall be fully inspection dimensionally to qualify a new or reworked cavity for use in production. The molded parts shall carry the individual cavity identification. As a control of each cavity during production, the above quantity of parts from each cavity shall be inspected for at least the defects listed in paragraphs 4.4.2.3 and 4.4.2.7 after continuous production of each 5,000 parts or at the end of each week, whichever occurs first. Of the three (3) samples, one (1) sample shall be the last part produced. If any defective parts are found during qualification of the cavity, the cavity producing the defective part will not be used in production. If any defective parts are found when inspection is performed for the control of the cavity, the cavity producing the defective part shall be removed from production. Further, that portion of production since the last control check shall be returned to the contractor for inspection for each separate type of defect according to MIL-STD-105, using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. All cavities removed from production because of some fault may, after reworking, be returned to production providing they pass the qualification test above. The contractor may request a change of inspection frequency providing he presents objective evidence to the Contracting Officer to substantiate the request. Contractor designs of gages and test equipment required to perform the inspections listed herein shall be forwarded to Commander, ARRADCOM, Dover, New Jersey 07801, ATTN: DRSAR-QA-I, for approval prior to manufacture of equipment. The noted sub-paragraphs identify those items and minimum inspections subject to the requirements of this paragraph.

NOTE 1: In establishing a cure time, dimensionally check ten (10) parts from each cavity at periodic intervals (e.g. every 30, 60, etc. mins) until dimensioanl stability is attained. The curing time will be from the time the part comes out of the mold until dimensional stability is attained. The inspection data in determining the curing time shall be sent to ARRADCOM, Dover, New Jersey 07801, ATTN: DRSAR-QAA-A. If there is a change in material, or in the cycle time, or if a cavity is reworked, or a new cavity is used, a new curing time shall be established and approved.

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4.5.2 Dimensional control of stamped parts. - In place of the normal sampling associated with the Classification of Defects, and with the approval of the Contracting Officer, a sample of at least ten (10) parts shall be dimensionally inspected to quality the tool used in the stamping process for use in production. In addition, a random sample of five (5) parts shall be selected from the last portion of each hour's production for dimensional inspection as a control of the tool during production.

If any defective parts are found during qualification of the tool, the tool producing the defective part shall not be used in produciton.

If any defective parts are found when inspection is performed for control of the tool, the tool producing the defective part shall be removed from production. Further, that portion of production since the last tool control check shall be returned to the contractor for corrective action.

All tools removed from production because of some fault, may, after reworking, be returned to production providing they pass the qualification test above. The contractor may request a change of inspection frequency providing he presents objective evidence to the Contracting Officer to substantiate the request.

4.5.3 Moisture content. -

4.5.2.1 Delay composition, smoke composition, ignition composition. - The moisture content of the compositions shall be sampled, at the time and place of loading, in accordance with Method 102.1 of MIL-STD-1234.

4.5.3.2 Disc, ejection composition. - The moisture content of the composition at the time and place of loading shall be determined in accordance with Paragraph 4.4.3 of MIL-P-223.

4.5.4 Salt spray. - This test shall be conducted in accordance with ASTM B 117-73.

4.5.5 Static burn time of delay loading assembly. - This test shall be conducted by any Government approved procedure.

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4.5.6 Static burn time of smoke canister loading assembly. - This test shall be conducted by any Government approved procedure.

4.5.7 Pull test of cartridge. - The cartridge shall be placed in an approved fixture and the axial force specified on the applicable drawing shall be applied. Cartridge shall be pulled until total separation occurs and data shall be recorded. This test is destructive test. Parts so tested shall not be returned to the lot.

4.5.8 Transportation vibration. - The cartridges shall be packaged and packed in accordance with dwgs. 9209204 and 9209205 and tested in accordance with Test NO. 104 of MIL-STD-331, Procedure II, except that each box shall be vibrated at the specified amplitudes for four (4) hours in each of three (3) different positions (i.e., box positioned so that the cartridges are vertical with case end down, box positioned so that the cartridges are horizontal, and box positioned so that the cartridges are vertical with case end up. After the test, the box packing and the cartridges shall be examined to determine compliance with the requirements. Cartridges shall be used for subsequent tests.

4.5.9 Air pressure. - The cartridge shall be placed in an approved fixture and a measured quantity of air shall be applied to produce the required air pressure. Cartridges shall be used for subsequent testing.

4.5.10 Functioning. - The following tests shall be performed at a Government-owned proving ground:

4.5.10.1 Hot temperature (160 degrees F. plus or minus 5 degrees F.) - The cartridges shall be packed in their regular shipping containers and conditioned for a minimum of 16 hours at a temperature of 160 degrees F. plus or minus 5 degrees F. The cartridge shall be taken from the conditioning chamber, removed from the packing container, and fired from an approved launcher placed on an approved mount at 90 degrees quadrant within approximately 2 minutes. Observation shall be made for proper functioning and the requirements of 3.4.5.

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4.5.10.2 Ambient temperature (70 degrees F. plus or minus 5 degrees F. - The cartridges shall be packed in their regular shipping containers and conditioned for a minimum of 16 hours at a temperature of 70 degrees F. plus or minus 5 degrees F. The cartridge shall be taken from the conditioning chamber, removed from the packing container, and fired from an approved launcher placed on an approved mount at 90 degrees quadrant within approximately 2 minutes. Observation shall be made for proper functioning and the requirements of 3.4.6.

4.5.10.3 Cold temperature (-65 degrees F. plus or minus 5 degrees F. - The cartridges shall be packed in their regular shipping containers and conditioned for a minimum of 16 hours at a temperature of -65 degrees F. plus or minus 5 degrees F. The cartridge shall be taken from the conditioning chamber, removed from the packing container, and fired from an approved launcher placed on an approved mount at 90 degrees quadrant within approximatley 2 minutes. Observation shall be made for proper functioning and the requirements of 3.4.7.

4.5.10.4 Test validity. - If for any reason the proving ground considers that the test conditions have detrimentally affected the test results, the test shall be declared invalid and a new test shall be performed with additional samples.

4.5.11 Air pressure of packed sealed metal box. - The sealed metal box shall be placed in an approved fixture and a measured quantity of air shall be applied to produce the required air pressure. Observation shall be made for leakage. This test is a non-destructive test. Boxes so tested may be returned tothe lot.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. -

5.1.1 Level A. - The cartridges shall be preserved and packaged in accordance with dwg. 9209204.

5.1.2 Level C. - The metal parts shall be preserved and packaged in accordance with MIL-STD-1169.

5.2 Packing. -

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5.2.1 Level A. - The cartridges, preserved and packaged as specified in 5.1.1, shall be packed in accordance with dwg. 9209205.

5.2.2 Level C. - The metal parts, preserved and packaged as specified in 5.1.2, shall be packed for shipment in accordance with MIL-STD-1169.

5.3 Marking. - Marking shall be in accordance with Dwgs. 9209204, 9209205, MIL-STD-1169 and Code of Federal Regulations, Title 49, Parts 100-199.

5.4 Shipping. - When shipments from more than one lot are shipped as a carload, each lot shall be kept separate and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Intended use. - This specification covers the metal parts and loading, assembling and packing for Cartridge: 40MM, White Smoke, Canopy, M680.

6.2 Ordering data. - See MIL-A-48078.

6.3 Submission of inspection equipment designs for approval. - See MIL-A-48078. Submit equipment designs as required to Commander, ATTN: DRDAR-QAA-I, ARRADCOM, Dover, New Jersey 07801.

6.4 Submission of test data. - In addition to the normal distribution of records, when the cartridge is procured by ARRCOM, one (1) copy of all ballistic data and ammunition data cards shall be forwarded to Commander, DRDAR-QAA-Q, ARRADCOM, Dover, New Jersey 07801.

6.5 The cartridges will be considered safe to transport providing no evidence exists of loose powder or composition in the box.

6.6 The cartridges will be considered free of damage that will affect the indented function provided the top seal has not been broken by movement or displacement of the top, or the case side has not been distorted sufficiently to prevent ejection of the illuminant assembly.

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6.7 Proving ground test. - Initial profuction testing shall be conducted in accordance withUS Army Test and Evaluation Command Initial Production Test Plan No. CART-40GL-VAR.

6.8 Drawings. - Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities are now under the cognizance of ARRADCOM.

6.9 Proving Ground Test Summary:

TEST	SAMPLE SIZE	REQUIREMENTS
Functioning		
(1) First Article		
a. Hot	80	See 3.4.5 & 4.3.3.1
b. Ambient	280	See 3.4.6 & 4.3.3.1
c. Cold	80	See 3.4.7 & 4.3.3.1
(2) First 3 lots		
a. Hot	44	See 3.4.5 & 4.4.3.5
b. Ambient	176	See 3.4.6 & 4.4.3.5
c. Cold	44	See 3.4.6 & 4.4.3.5
(3) Regular Production		
a. Hot	26	See 3.4.5 & 4.4.3.8
b. Ambient	80	See 3.4.6 & 4.4.3.8
c. Cold	26	See 3.4.7 & 4.4.3.8

Custodian:
Army-AR

Preparing Activity:
Army-AR

Project Number: 1310-A-096

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